

5(a)	crossing over ; A chiasma formation	1
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Question	Answer	Marks
5(b)(i)	any three from: 1 mutation occurred a long time ago or someone in original population had mutation or mutation in common ancestor ; 2 <i>ref. to.</i> founder effect ; 3 (original / current) population is small ; 4 <i>idea that</i> much of the population is closely related or small gene pool / low genetic diversity ; A inbreeding 5 <i>ref. to</i> isolated (population) ; 6 (caused) increase in frequency of, mutation / 999del5 / <i>BRCA2</i> allele ;	3
5(b)(ii)	any three from: 1 reduce worry if negative ; (if positive) 2 lifestyle changes / described ; 3 early treatment / breast removal / can have frequent checks ; 4 informed decision about having children ; 5 <i>ref. to</i> counselling ; 6 could lead to lower death rates from breast cancer ;	3

Question	Answer	Marks
5(b)(iii)	any one from: 1 only screen for mutations that occur with high frequency in a population (as opposed to hundreds of mutations) ; 2 cost effective ; 3 early diagnosis / early treatment / targeted treatment ; 4 identifies mutation known to increase the risk of developing breast cancer (as opposed to a neutral mutation) ; 5 could lead to lower death rates from breast cancer ;	1
5(c)(i)	any two from: 1 it, inserted / deleted / replaced, DNA (in, <i>Lcn2</i> / regulatory sequence) ; 2 (so) transcription of, <i>Lcn2</i> / gene, prevented ; 3 (replaced mutated section of DNA) with correct sequence ;	2
5(c)(ii)	only specific, sites / sequences, (on DNA targeted) or target site on DNA only found in <i>Lcn2</i> ;	1